Exciton torque

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Useful references and info about the exciton in tj model

I. USEFUL REFERENCES

[1] They compute exciton spectra and optical conductivity using t-J model on a four-leg cylinder using MPS (Their model is the typical t-j model with Hubbard and J between nearest neighbors), starting from half-filling ground state and applying an exciton operator, where, depending on the dopant (fermionic, bosonic, or a mixture) the corresponding excitonic operator is defined. More details about the MPS simulations are in Appendix C. Their ultimate goal is to compare against the Fermi-Hubbard and understand the nature of their excitations, e.g., exciton vs distinguishable dopants, allowing the tracking of the nature of the Fermi-Hubbard excitons.

II. MODEL AND METHOD

We consider an effective electron-hole model with spin degrees of freedom coupled to localized semiclassical spins

$$\hat{H} = \hat{H}_0 + \hat{H}_{sd}[\mathbf{M}(t)] + \hat{H}_U + \hat{H}_V \tag{1}$$

$$\hat{H}_0 = \sum_{\langle ij \rangle, l} \hat{c}_i^{l\dagger} H_{0,ij}^l \hat{c}_j^l \tag{2}$$

$$\hat{H}_U = U \sum_{il} \hat{n}_{i\uparrow}^l \hat{n}_{i\downarrow}^l \tag{3}$$

$$\hat{H}_V = -V \sum_{\sigma\sigma' l} \hat{n}_{i\sigma}^l \hat{n}_{i\sigma'}^l \tag{4}$$

$$\hat{H}_{sd}(t) = -J_{sd} \sum_{i,l} \mathbf{M}_i^l(t) \cdot \hat{\mathbf{s}}_i^l \tag{5}$$

The standard semiclassical dynamics for the LMMs coupled to electrons is obtained as a saddle point of the Gaussian approximation of the effective action (once the electrons are integrated out), which only works at small coupling $j_{sd} \ll \gamma$, obtaining an effective nonlocal damping. In this approximation, the Kernel is obtained from the magnetic susceptibility of the electrons:

$$R_{ij}^{ab}(t) = J_{sd}^2 \int_{-\infty}^{t} dt' \chi_{ij}^{ab,R}(t-t') \tag{6}$$

$$\chi_{ij}^{ab,R}(t,t') = -i\theta(t-t')\langle [\hat{s}_i^a(t), \hat{s}_i^b(t')] \rangle \tag{7}$$

Thus by exact methods, it is possible to get full suceptibility. On the other hand, once the excitonic effects are included...

The bare dynamics of $\mathbf{M}_i^l(t)$ is described by the classical hamiltonian $H_{cl}[\mathbf{M}]$. Our goal is to derive an effective semiclassical dynamics for the LMMs, including the effects of correlated electrons within each layer and exciton formation in between layers.

$$\partial_t \mathbf{M}^l(t) = \mathbf{M}^l(t) \times \Big[\mathbf{B}_{cl}^l[\{\mathbf{M}(t)\}] + \mathbf{B}_\Phi^l[\{\mathbf{M}(t)\}] + \mathbf{B}_0^l[\{\mathbf{M}(t)\}] \Big] \tag{8}$$

$$\mathbf{B}^{l,cl}[\mathbf{M}(t)] = -\frac{\partial H^{cl}[\{\mathbf{M}(t)\}]}{\partial \mathbf{M}^l(t)} \tag{9}$$

A. Excitonic torque ($U = 0$)

We write the effective action as a function of the intralayer excitonic order parameter $\hat{B}_i^{\sigma\sigma'} = \hat{c}_{i\sigma}^{1\dagger} \hat{c}_{i\sigma'}^2$, using $\hat{n}_i^1 \hat{n}_i^2 = 2\hat{n}_i^1 - \sum_{\sigma,\sigma'} \hat{B}_i^{\sigma\sigma'} \hat{B}_i^{\sigma\sigma'}$ and absorbing the first term into the chemical potential, the electronic action $\mathcal{S}_e$ reads

$$\mathcal{S}_e = \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}_i^T G_{\mathbf{M}}^{-1} \Psi_j + V \int_{\mathcal{C}} dz \sum_{i\nu} B_i^\nu \bar{B}_i^\nu, \tag{10}$$

where $\Psi = (c_\uparrow^1, c_\downarrow^1, c_\uparrow^2, c_\downarrow^2)^T$, and the polarized Green function (GF) satisfies $\big(i\partial_z - H_0 - H_{sd}[\mathbf{M}]\big) G_{\mathbf{M}}(z,z') = \delta_C(z,z')$. We additionally decompose the excitonic field on its charge and spin degrees of freedom using $B_i^\nu = \mathrm{Tr}_\sigma[\Gamma^\nu B_i]$, with $\Gamma^\nu = \sigma^\nu/\sqrt{2}$ and $\nu = 0, x, y, z$. Applying the Hubbard Strattonovich (HS) transformation $\mathcal{S}_V \to \mathcal{S}_V^{HS}[\{\Phi_i^\nu\}]$, the action is written as

$$\begin{aligned}\mathcal{S}_e &= \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}_i^T (G_{\mathbf{M},ij}^{-1} - \delta_{ij} \mathcal{V}_\Phi)_{ij} \Psi_j \\ &+ \frac{1}{V} \int_{\mathcal{C}} dz \sum_{i,\nu} \bar{\Phi}_{\mathrm{i}}^\nu \Phi_{\mathrm{i}}^\nu. \end{aligned} \tag{11}$$

Integrating out the electrons and keeping to second order (RPA), the effective exciton action reads.

$$\mathcal{S}^{\rm eff}_{\Phi}[\mathbf{M}] = \int_{\mathcal{C}} dzdz' \sum_{\nu\nu'} \bar{\Phi}^{\nu} \mathrm{D}^{-1}_{\nu\nu'} \Phi^{\nu'} - \mathrm{iTrln}[\mathrm{G}^{-1}_{\mathbf{M}}] \quad (12)$$

where we assumed a homogeneous order parameter for the exciton, the propagator of the exciton satisfies the Bethe-Salpeter equation $D = V(1 + \Pi D)$

$$D^{-1}_{\mu\nu}(z,z') = \frac{\delta_{\mathcal{C}}(z,z')\delta_{\mu\nu}}{V} - \Pi_{\mu\nu}(z,z') \quad (13)$$

$$\Pi_{\mu\nu}(z,z') = -\frac{i}{L}\mathrm{Tr}[\Gamma^{\mu\dagger} G^1_{\mathbf{M}}(z,z')\Gamma^{\nu} G^2_{\mathbf{M}}(z',z)], (14)$$

Additionally, we integrate out the exciton degrees of freedom to get an effective action for the classical moments. The total effective action reads,

$$\mathcal{S}^{\rm eff}_{\rm cl} = \mathcal{S}_{cl} - i\mathrm{Trln}[G^{-1}_{\mathbf{M}}] + i\mathrm{Trln}[D^{-1}] \quad (15)$$

In order to find the semiclassical dynamics, we minimize with respect to the classical LLMs $\frac{\partial \mathcal{S}^{\rm eff}_{\rm cl}}{\partial \mathbf{M}^l} = 0$, the classical term, which contains the Berry phase and its respective classical Hamiltonian of the LMMs returns

$$\frac{\partial \mathcal{S}_{cl}}{\partial \mathbf{M}^l(t)} = \frac{1}{\gamma_l}\mathbf{M}^l \times \partial_t \mathbf{M}^l + \mathbf{B}^{l,cl} \quad (16)$$

The contributions to the effective field from the polarized electrons is

$$\begin{aligned}\mathbf{B}^l_0[\{\mathbf{M}(t)\}] &= i\frac{\partial \mathrm{Trln}[G^{-1}_{\mathbf{M}}]}{\partial \mathbf{M}^l} \\ &= J_{sd}\langle \hat{\mathbf{s}}^l \rangle_{\mathbf{M}} \quad (17)\end{aligned}$$

and the contribution from the exciton is

$$\begin{aligned}\mathbf{B}^l_{\Phi}[\{\mathbf{M}(t)\} &= -i\frac{\partial \mathrm{Trln}[D^{-1}]}{\partial \mathbf{M}^l} \\ &= -i\mathrm{Tr}\Big[D\frac{\partial \Pi}{\partial \mathbf{M}^l}\Big] \quad (18)\end{aligned}$$

Where the explicit form of the torques is: (To compute the real-time dynamics, we should apply Langreth rules to get the retarded component):

$$B^1_{\alpha\Phi}(\bar{z}) = \int dzdz' \frac{j_{sd}}{L} \sum_{\mu\nu} D_{\mu\nu}(z,z') \mathrm{Tr}\big[\Gamma^{\mu} G^2_{\mathbf{M}}(z',\bar{z})\Gamma^{\nu} G^1_{\mathbf{M}}(\bar{z},\bar{z}')\sigma^{\alpha} G^1_{\mathbf{M}}(z',z)\big] \quad (19)$$

$$B^2_{\alpha\Phi}(\bar{z}) = \int dzdz' \frac{j_{sd}}{L} \sum_{\mu\nu} D_{\mu\nu}(z,z') \mathrm{Tr}\big[\Gamma^{\mu} G^2_{\mathbf{M}}(z',\bar{z})\sigma^{\alpha} G^2_{\mathbf{M}}(\bar{z},z)\Gamma^{\nu} G^1_{\mathbf{M}}(z,z')\big]$$

Putting all together, and following the standard procedures, we finally arrive to Eq. [(**??**)]

Up to this point, the theory is completely self-consistent, $\mathbf{M} \to G_{\mathbf{M}} \to D \to \mathbf{M}$. At each time step, the classical magnetization, the polarized GF, and the exciton propagator should be evolved self-consistently. There are two paths that can be followed:

- Given a classical trajectory of the LMMs, calculate the exciton response and spectra from $D[\mathbf{M}]$
- Use a j_{sd} expansion to get explicit expressions for torques without self-consistency

1. j_{sd} *expansion and LLG equations*

In order to get the standard LLG, the effective fields should be expanded to first order in M. This can be done in two different ways:

- Considering and adiabatic approach, where $G_{\mathbf{M}}$ is computed explicitly from a frozen configuration. (It sounds like Born-Oppenheimer nucleus-electron motion), and still evolve self-consistently
- Expanding to first order $G_{\mathbf{M}} = j_{sd}\mathbf{M}G^{(1)} + O(j^2_{sd})$ and eliminate the selfconsistency because the kernels can be computes independently

The second way emerges naturally as a second-order expansion of S^{eff}, and is the natural way to get the effective LLG equation type. In our case, the susceptibility comes from

$$(j_{sd})^2\chi^{ll'}_{ab}(z,z') = \frac{\partial \mathcal{S}^{\rm eff}}{\partial M^l_a(z)\partial M^{l'}_b(z')} \quad (20)$$

For the electronic mediated kernel, the standard non-local kernel [2], where only the retarded component contributes to the equations

$$R^{0,ll'}_{ab}(z,z') = (j_{sd})^2\delta_{ll'}\sum_k \mathrm{Tr}[\sigma^a g^l_k(z,z')\sigma^b g^l_k(z',z)] (21)$$

$$= (j_{sd})^2\delta_{ll'}\delta_{ab}\sum_k g^l_k(z,z')g^l_k(z',z) (22)$$

The susceptibility contribution from the exciton is:

$$R_{ab}^{\Phi(I),ll'}(z,z') = -i\mathrm{Tr}_\nu[D^0 \cdot \Lambda_{ab}^{ll'}](z,z') \quad (23)$$

$$R_{ab}^{\Phi(II),ll'}(z,z') = i\mathrm{Tr}_\nu[D^0 \cdot \Lambda_l^a(z) \cdot D^0 \cdot \Lambda_{l'}^b(z')] \quad (24)$$

where the vertex functions $\Lambda_a^l(z,z';\bar{z}) = \frac{\partial \Pi_\mu(z,z')}{\partial M_a^l(\bar{z})}$ are:

$$\Lambda_a^1(z,z';\bar{z}) = i\frac{J_{sd}}{L}\sum_k \mathrm{Tr}\big[\Gamma^\mu G_{\mathbf{M}}^2(z,z')\Gamma^\nu G_{\mathbf{M}}^1(z',\bar{z})\sigma^a G_{\mathbf{M}}^1(\bar{z},z)\big]$$

$$\Lambda_a^2(z,z';\bar{z}) = i\frac{J_{sd}}{L}\sum_k \mathrm{Tr}\big[\Gamma^\mu G_{\mathbf{M}}^2(z,z')\sigma^a G_{\mathbf{M}}^2(z',\bar{z})\Gamma^\nu G_{\mathbf{M}}^1(\bar{z},z)\big]$$

fot the 2 vertexes we got

$$\Lambda_{ab}^{21}(z,z';\bar{z},\bar{z}') = -i\frac{j_{sd}^2}{L}\mathrm{Tr}\Big[\Gamma^\nu G_{\mathbf{M}}^2(z,\bar{z})\sigma_a G_{\mathbf{M}}^2(\bar{z},z')\Gamma^\nu G_{\mathbf{M}}^1(z',\bar{z}')\sigma_b G_{\mathbf{M}}^1(\bar{z}',z)\Big] \quad (25)$$

the

2. Adiabatic expansion, damping, inertia and so ...

To derive explicit non-local...

$\Lambda_a^l(z')$ denotes the 1 vertexes

B. Semiclassical Low U Effective Theory $(U, V \ll \gamma)$

$$\mathcal{S}_V = V\int_{\mathcal{C}} dz \sum_{i\sigma\sigma'} B_i^{\sigma\sigma'}\bar{B}_i^{\sigma\sigma'} \quad (26)$$

C. Semiclassical Strong U Effective Theory or Bilayer tJ Model $(U \gg \gamma)$

[1] A. Bohrdt, E. Demler, and F. Grusdt, Spectroscopy of hubbard-mott excitons and their ro-vibrational excitations, arXiv preprint arXiv:2406.16854 (2024).

[2] F. Reyes-Osorio and B. K. Nikolić, Gilbert damping in metallic ferromagnets from schwinger-keldysh field theory: Intrinsically nonlocal, nonuniform, and made anisotropic by spin-orbit coupling, Phys. Rev. B **109**, 024413 (2024).